



Contents lists available at ScienceDirect

Journal of Global Antimicrobial Resistance

journal homepage: www.elsevier.com/locate/jgar

Short Communication

Successful use of cefepime/zidebactam in deep-seated infections due to extensively drug-resistant *Pseudomonas aeruginosa*: A case seriesPooja Shah^a, Ayyanraj Neeravi^b, Rajiv Karthik^a, Christy Merin Mathew^a, Julie Hephzibah^c, Hanna Elizabeth Johnson^c, Balaji Veeraraghavan^b, Abi Manesh^{a,*}^a Department of Infectious Diseases Training and Research Centre, Christian Medical College, Vellore, Tamil Nadu, India^b Department of Clinical Microbiology, Christian Medical College, Vellore, Tamil Nadu, India^c Department of Nuclear Medicine, Christian Medical College, Vellore, Tamil Nadu, India

ARTICLE INFO

Article history:

Received 8 December 2024

Revised 14 June 2025

Accepted 7 July 2025

Available online xxx

Editor: Minggui Wang

Keywords:

Cefepime/zidebactam

B-lactam enhancer

Pseudomonas aeruginosa

Extensively drug resistant

ABSTRACT

Objective: There is a paucity of safe and effective treatment options for extensively drug resistant (XDR) Gram-negative pathogens that often harbor varied resistance mechanisms concurrently. Cefepime/zidebactam (FEP/ZID) is a novel β -lactam/ β -lactam enhancer (BL-BLE) combination, currently in Phase 3 trials, that has been shown to be effective against such difficult to treat resistant isolates in vitro studies and scattered case reports. We present a series of deep seated infections due to XDR *Pseudomonas Aeruginosa* (PsA), successfully treated with FEP/ZID under compassionate use.

Methods: Between September 2023 and May 2024, using clinico-radiographic and microbiologic (phenotypic drug susceptibility and whole genome sequencing) data, we identified five XDR PsA related infections that could not be safely treated with currently available antibiotics. All patients received FEP/ZID after obtaining informed consent and necessary approvals.

Results: Four cases involved osteoarticular disease while the fifth was an endovascular graft infection. Clinical specimens yielded XDR PsA isolates that harbored varied (enzymatic and non-enzymatic) resistance mechanisms to anti-pseudomonal cephalosporins, carbapenems, aminoglycosides, fluoroquinolones and newer β -lactam/ β -lactamase inhibitors. All isolates were susceptible to FEP/ZID and were treated with the same for 2–6 wk. Clinico-radiographic assessments confirmed significant resolution of lesions in all cases. There were with no drug-related adverse events.

Conclusions: The in-vitro susceptibility of our XDR PsA isolates to FEP/ZID translated to a successful clinical outcome in our series of deep-seated infections. FEP/ZID is a potential safe and effective treatment option for such serious Gram-negative infections.

© 2025 The Authors. Published by Elsevier Ltd on behalf of International Society for Antimicrobial Chemotherapy. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

1. Introduction

Infections caused by *Pseudomonas Aeruginosa* (PsA) are often complicated by multi-drug resistance and contribute to half a million global deaths annually [1]. Consequently, carbapenem-

resistant PsA has been assigned a high-priority status on the 2024 WHO priority list of bacterial pathogens that urgently require newer antibiotics [2]. This unmet need for effective antibiotics is most pronounced for XDR PsA isolates that are susceptible to two or fewer classes of antibiotics [3] and often harbor both enzymatic and non-enzymatic mechanisms of resistance concurrently. The co-existence of broad-spectrum β -lactamases (particularly cephalosporinases and metallo-beta lactamases-MBLs) along with over-expression of efflux, loss of porins and/or target site alterations – in varying permutations – limit the utility of newer β -lactam/ β -lactam inhibitor (BL-BLI) combinations like aztreonam/avibactam (AZT/AVI), ceftolozane/tazobactam (TOL/TAZO) and imipenem/relebactam (IMI/REL)[4]. This often results in the use of agents with poor efficacy and safety, typically polymyxins.

Abbreviations: OXA, Oxacillinases; VEB, Vietnamese Extended-spectrum Beta-lactamase; Mex, multidrug efflux; Opr, Outer-membrane porin; PAC, *Pseudomonas Aeruginosa* Class C; PDC, *Pseudomonas-Derived* Cephalosporinase; IMP, Imipenemase; VIM, Verona Integron-encoded Metallo-beta lactamase; PER, *Pseudomonas* Extended Resistance; GES, Guiana Extended Spectrum Beta Lactamase; piuA, Pseudomonas iron uptake protein A.

* Corresponding author: Department of Infectious Diseases Training and Research Centre, Christian Medical College, Ida Scudder Road, Vellore, Tamil Nadu 632004, India.

E-mail address: abimanes@gmail.com (A. Manesh).

<https://doi.org/10.1016/j.jgar.2025.07.009>

2213-7165/© 2025 The Authors. Published by Elsevier Ltd on behalf of International Society for Antimicrobial Chemotherapy. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Please cite this article as: P. Shah, A. Neeravi, R. Karthik et al., Successful use of cefepime/zidebactam in deep-seated infections due to extensively drug-resistant *Pseudomonas aeruginosa*: A case series, Journal of Global Antimicrobial Resistance, <https://doi.org/10.1016/j.jgar.2025.07.009>

FEP/ZID is a novel BL-BLE combination that remains active against a multitude of enzymatic (including MBLs) and non-enzymatic resistance mechanisms. This is due to its unique, simultaneous and reciprocal inhibition of multiple penicillin binding proteins (PBPs) – namely PBP-2 by zidebactam and PBP-1a and 3 by cefepime. Its broad antibacterial spectrum, coupled with favorable pharmacokinetic features [5], has resulted in scattered reports of its compassionate use [6–9]. However, its efficacy and safety in deep-seated infections, often requiring prolonged therapy, is largely unknown. We present a series of five patients with such infections, caused by XDR PsA, that were successfully treated with FEP/ZID.

2. Methodology

We treated five patients with XDR PsA related infections using FEP/ZID in our hospital between September 2023 and May 2024. Microbiological susceptibility testing for each of the isolates was done (in duplicates) using standardised techniques (supplementary material). Antibiotic susceptibilities were interpreted according to the Clinical and Laboratory Standards Institute (CLSI) 2024 criteria. With regards to FEP/ZID, the CLSI has recently assigned an investigational breakpoint of ≤ 64 mg/L for PsA [10]. Phenotypic susceptibility testing was then correlated with Whole Genome Sequencing (WGS) performed using standard methods (supplementary material).

With this microbiological and concurrent clinico-radiographic information, it was confirmed that all five patients had serious infections that could not be treated with currently available treatment options. The decision to use FEP/ZID was made by a team of infectious diseases physicians (in conjunction with the hospital clinical ethics committee) and was followed by written informed consent of the concerned patients. The Institutional Review Board also approved the study (IRB number: 0,424,159). FEP/ZID injection vials were supplied by Wockhardt Ltd after approval from the Drug Controller General of India (DCGI).

3. Case reports

Table 1 provides the comparative clinico-radiographic profile, microbiological diagnosis, treatment details and outcomes of the five patients treated with FEP/ZID

3.1. Case 1

A 72-y-old gentleman with multiple comorbidities (Parkinson's disease, benign prostatic hypertrophy and hypertension), presented with left-sided ear pain and headache since two months followed by ipsilateral lower motor neuron (LMN) type of facial palsy since 3 wk. Left-sided malignant otitis externa (MOE) and lateral skull base osteomyelitis (SBO) were diagnosed by clinical evaluation, Gallium 68-citrate Positron Emission Tomography Computed Tomography (Ga-68 citrate PET-CT) scan and magnetic resonance imaging (MRI) of the brain and skull base. Intra-operative cultures (from condylectomy) yielded XDR PsA with compatible biopsy findings of pyogenic osteomyelitis. Anti-pseudomonal cephalosporins (ceftazidime and cefepime), BL-BLIs (piperacillin-tazobactam and cefoperazone-sulbactam), carbapenems (mero- and imipenem), levofloxacin and amikacin were not susceptible (Table 2). As per WGS data, resistance to levofloxacin was mediated by target site mutations in deoxyribonucleic acid (DNA) gyrase A and topoisomerase IV, while that to aminoglycosides was attributable to inactivating enzymes – ribosomal methyltransferase (RMTase) B and aminoglycoside modifying enzymes (AMEs). With regards to beta lactam resistance, there were class D β -lactamases (OXA-846 & OXA-10) and cephalosporin-hydrolysing

enzymes ($\text{bla}_{\text{VEB-9}}$ & $\text{bla}_{\text{PDC-11}}$). As there were no carbapenemase or porin encoding mutations noted, resistance to carbapenems and BL-BLIs (ceftazidime/avibactam (CAZ/AVI), AZT/AVI, IMI/REL and TOL/TAZO) was likely caused by the up-regulation of efflux pumps, specifically mexAB-OprM (supplementary material, Table 1). Given the limited bone penetration and potential nephrotoxicity with colistin, FEP/ZID was initiated for a duration of 6 wk. At the end of treatment, the patient reported complete alleviation of headache and was noted to have normalisation of Capsular Reactive Protein (CRP) from the initial value of 115 mg/L. The only minor, probable drug related adverse event was a transient drop in platelet count (from a peak value of $215 \times 10^9/\text{L}$ to a nadir of $103 \times 10^9/\text{L}$). Counts were monitored closely without any therapy modification and recovered completely after treatment conclusion. A regional PET-CT at 3 months showed complete resolution of uptake in the previously affected areas.

3.2. Case 2

A 59-y-old diabetic gentleman reported a history of treated left proximal tibial chronic osteomyelitis following internal fixation for a traumatic open fracture in 2018. He was well till December 2023, when he presented to us with acute left knee infective arthritis. Synovial fluid cultures (from joint debridement) isolated XDR PsA with corroborative pyogenic arthritis on bone biopsy. The resistance type was similar to the previous isolate (ST357; Table 2). Notable differences in WGS data were varied class D β -lactamases (OXA-796, OXA-50, and OXA-10) and cephalosporin-hydrolysing enzymes ($\text{bla}_{\text{PAC-1}}$ and $\text{bla}_{\text{PDC-98}}$) along with mutations in PBP-3 (supplementary material, Table 1). Within 5 d of polymyxin based therapy, he developed drug induced perioral paresthesia and progressive acute kidney injury (AKI). Consequently, it was discontinued and FEP/ZID was initiated for 4 wk. At the end of treatment, he showed a remarkable improvement in ambulation and normalisation of CRP from the initial elevated value of 165 mg/L. Ga-68 citrate scan at this point showed a significant decrease in joint uptake and effusion. No drug-related adverse events (ADE) were recorded. He remained pain-free at 9-month follow-up.

3.3. Case 3

A 54-y-old gentleman, with poorly controlled hypertension, underwent an emergency repair of a ruptured infra-renal atherosclerotic aortic aneurysm in August 2022. His post-operative course was complicated by an infected pseudoaneurysm at the anastomotic site of the aorta with the right femoral artery. This was managed with excision of the aneurysm and re-anastomosis of the graft to the superficial femoral artery. The intra-operative tissue cultures isolated carbapenem-resistant PsA with synergy for CAZ-AVI + AZT. He was treated (elsewhere) with the same for 14 d followed by AZT monotherapy for 4 wk (rationale unclear). In May 2023, he presented to us with an infected collection at the anastomotic site and underwent an extra-anatomical obturator bypass with excision of the infected graft. Intra-operative pus cultures isolated XDR PsA. The patterns and mechanisms of resistance were similar to the isolate seen in case 1 (ST357) except for evidence of New Delhi Metallo- β -lactamase (NDM) on Xpert Carba-R (supplementary material, Table 1) and notable synergy for AZT/AVI (Table 2). However, despite 2 wk of therapy with CAZ-AVI + AZT, fever persisted and CRP increased from 21 to 40 mg/dL. Considering this poor response, FEP/ZID was initiated for 14 d. This was followed by sustained defervescence, cessation of sinus discharge and normalisation of CRP. There were no ADE noted. Fludeoxyglucose-18 PET-CT (at 6 wk postop) showed marked reduction in the peri-graft collection. At 6-month follow-up, he remained asymptomatic with a

Table 1

Summary of clinico-radiographic diagnosis, patient profile, microbiologic etiology, treatment details and outcomes.

Case	Diagnosis	Co-morbidities	Organism isolated	Duration of FEP/ZID treatment (Rx) (in weeks)	Self-reported pain score*		C-reactive protein (CRP) (in mg/L)		Comparative PET-CT findings- before and after Rx	Clinical outcome
					Before Rx	After Rx	Before Rx	After Rx		
1	Left-sided MOE & lateral SBO	Parkinson's disease, hypertension, benign prostatic hypertrophy	<i>Extensively drug resistant (XDR)- P. aeruginosa (PsA)</i>	6	9/10	0/10	115	Normal**	Near resolution of gallium-68 (Ga-68) citrate avid uptake in the left external auditory canal and adjacent skull base	Cured without any evidence of active disease on follow-up
2	Left knee infective arthritis	Diabetes mellitus	<i>XDR-PsA</i>	4	10/10	0/10	165	Normal	Significant decrease in left knee joint uptake and effusion	Cured without any evidence of active disease on follow-up
3	Aorto-femoral anastomotic site infection	Hypertension & pre-diabetes	<i>XDR-PsA</i>	2	3/10	0/10	40	Normal	No 18-FDG avid uptake at aorto-femoral vascular graft site	Cured without any evidence of active disease on follow-up
4	Right-sided MOE, lateral + central SBO & temporal lobe leptomeningitis	Diabetes mellitus, congestive heart failure and chronic liver disease	<i>XDR-PsA</i>	4	9/10	2/10	8.3	Normal	Not done	Cured
5	Left-sided MOE with lateral & central SBO	Hypertension & diabetes mellitus	<i>XDR-PsA</i>	4	9/10	0/10	146	Normal	Near resolution of the Ga-68 citrate avid uptake in the left external auditory canal and adjacent skull base	Cured

Self-reported pain score was assessed as per the standardised Numerical Rating Scale (NRS) Normal CRP was defined as <6 mg/L.

Table 2

In vitro activity of FEP/ZID and comparator anti-pseudomonal antibiotics against clinical isolates.

Antibiotics	MIC (mg/L)				
	Case 1	Case 2	Case 3	Case 4	Case 5
Cefepime	>128	>128	>128	>128	>128
Zidebactam/cefepime (1:1)	32	2	2	4	8
Ceftazidime/avibactam (4 mcg)	>128	>128	>128	>128	>128
Ceftolozane/tazobactam	>128	128	>128	>128	>128
Imipenem	>128	16	128	8	>128
Imipenem/relebactam (4 mcg)	>128	8	128	8	>128
Meropenem	>128	>128	128	64	>128
Meropenem/EDTA (200 mcg)	16	128	0.12	1	32
Aztreonam/avibactam (4 mcg)	64	64	4	32	64
Colistin	0.25	<0.12	0.25	<0.12	<0.12
Levofloxacin	>128	8	>128	32	>128
Amikacin	128	>128	128	128	128

MICs of antibiotics were determined by reference broth microdilution recommended by CLSI M07-Ed11. Zidebactam/cefepime combination was tested at 1:1 ratio as recommended by CLSI. Fixed concentration of 4 mg/L was used for avibactam, tazobactam and relebactam. EDTA: ethylenediamine tetraacetic acid was combined at fixed concentration of 200 mg/L.

persistently normal CRP and near-complete resolution of the aforementioned collection on PET imaging (Fig. 1).

3.4. Case 4

A 55-y-old gentleman, with multiple comorbidities (diabetes mellitus, cardiac failure and chronic liver disease), presented with a four-month history of right ear pain followed by ipsilateral LMN type of facial palsy. Three months prior to the current presentation, he was empirically treated with cefpodoxime-clavulanic acid for 1 wk followed by cefoperazone-sulbactam (1.5 grams 12 hourly) for the next week. This was followed by oral levofloxacin (500 mg once daily) for 15 d. MRCT (magnetic resonance and computed tomography) imaging of the brain and skull base revealed

features of right-sided MOE, contiguous SBO and temporal lobe leptomeningitis. Intra-operative cultures (from nasopharyngectomy and condylectomy) isolated XDR PsA with a similar resistotype as seen in case 2 (ST664; Table 2). Additionally, WGS detected NDM-1 (supplementary material, Table 1). Within 4 d of treatment with polymyxin B, he developed AKI and pulmonary edema. Thus, polymyxin was discontinued and FEP/ZID was initiated for 4 wk. At the end of therapy, he had complete resolution of headache and normalisation of CRP, without any ADE.

3.5. Case 5

A 62-y-old gentleman with multiple co-morbidities (diabetes mellitus, hypertension and previous stroke) presented to us with

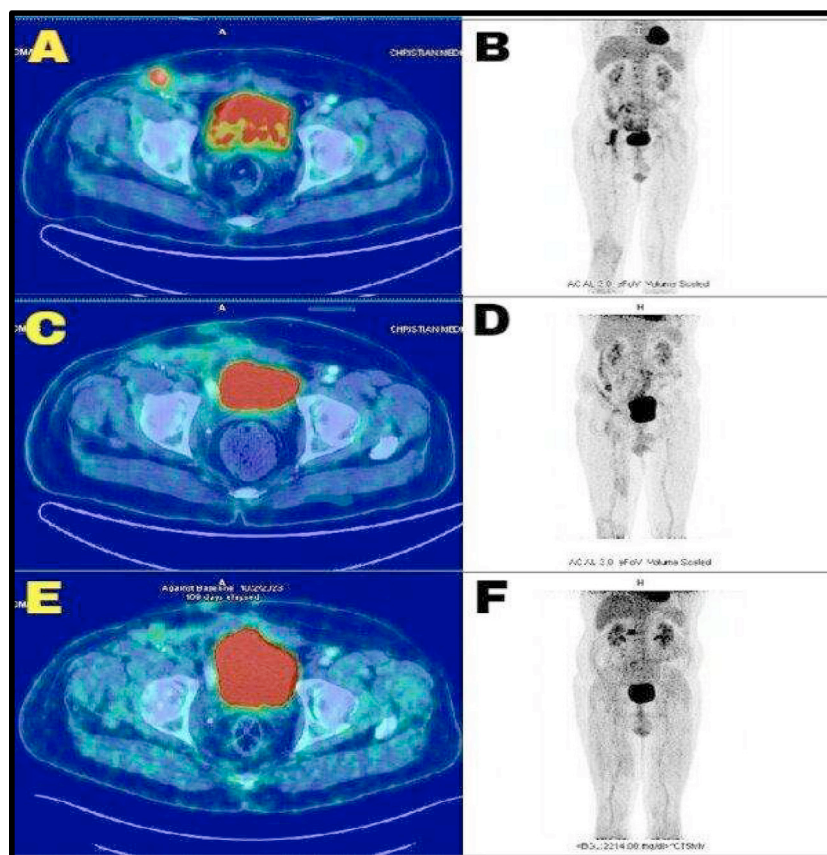


Fig. 1. 18-FDG PET images in case 3. (A and B) Pre-treatment images: soft tissue thickening and fluid collection around the distal right external iliac artery and common femoral artery. (C and D) Post-treatment with FEP-ZID: resolution of the previous fluid collection and stable soft tissue thickening with significant decrease its FDG-avidity. E-F: At 6 months follow up: Decrease in the extent of the peri-graft soft tissue thickening and near-resolution of FDG avidity.

left-sided MOE with contiguous lateral and central SBO. Intra-operative cultures (from condylectomy and nasopharyngectomy) yielded XDR- PsA with a similar resistotype as seen in case 3 (ST773). WGS could not be performed on this isolate; instead, multiplex polymerase chain reaction (PCR) was done which revealed the presence of NDM. However, there was no synergy for CAZ-AVI + AZT (Table 2), likely due to presence of efflux pumps. Following 2 wk of polymyxin B, he reported persistent headache and was noted to have a progressive AKI. Thus, FEP/ZID was initiated on a compassionate basis. After 4 wk of treatment, there was complete resolution of headache and normalisation of CRP (from the initial value of 146 mg/L) without any ADE. Follow-up Ga-68 citrate scan at this point showed a remarkable decrease in the enhancement over the affected skull base.

4. Discussion

Global collections of PsA are known to harbor diverse resistance mechanisms. As highlighted by the 2024 IDSA guidance on the treatment of multi-drug resistant Gram-negative infections, the relative prevalence of each of these resistance mechanisms and therefore the therapeutic options vary widely across different geographic settings [11]. In India, production of cephalosporinases and carbapenamases (particularly MBLs), contributes significantly to the beta lactam resistance seen in PsA [12]. Though AVI/AZT is stable against MBLs, the frequent co-existence of hyper-efflux precludes the widespread use of this combination; TOL/TAZO and IMI/REL have no activity against MBLs while cefepime-taniborbactam is ineffective against their IMP subtype

[4]. Over-expression/mutated forms of cephalosporinases and extended spectrum OXA beta-lactamases along with mutations in porins and PBPs are the other mechanisms of resistance against these newer BL-BLIs [4].

Though cefiderocol has potent activity against difficult to treat resistant (DTR) PsA, higher MICs are noted for isolates harboring certain beta-lactamases (PER, PDC-30, and GES-6), particularly specific types of MBLs (VIM-2, NDM-1, 5, 7, and 9). It is also ineffective in isolates with mutations affecting siderophores (piuD), over-expression of efflux pumps (MexAB-OprM) and decrease in porin channels (OprD) [13].

Zidebactam is a novel bicycloacyl-hydrazide DBO (diazabicyclooctane) uniformly stable against all β -lactamases including MBLs. It also exhibits intrinsic antibacterial activity due to its high-affinity binding to PBP-2. When used in conjunction with cefepime that targets PBP-1a and PBP-3, the combination produces a potent bactericidal effect due to its rapid, multi-site and complementary PBP inhibition. Unlike many other BL-BLIs, its antibacterial activity is dependent on concomitant saturation of multiple PBPs rather than the inhibition of the β -lactamases [5]. Therefore, it overcomes resistance that is mediated by carbapenamases as well as by non-enzymatic resistance mechanisms. This is particularly relevant for the XDR PsA isolates seen in India, which often harbor multiple resistance mechanisms concurrently [12]. Such a resistotype was noted in a majority of the isolates in our study as well (supplementary material, Table 2).

Since its discovery, FEP/ZID has undergone several phase 1 studies (NCT02674347, NCT02707107, and NCT03554304) [14–16] that have established its safety. It was also assessed (NCT02942810) [17]

in individuals with altered kidney functions for dose determination in renal insufficiency. A global, randomised, double-blind, multicenter, non-inferiority, phase 3 study (NCT04979806) [18] to evaluate its efficacy and safety in complicated urinary tract infections (cUTIs) is currently underway.

In India, it is presently undergoing a phase 2 observational study (CTRI/2024/04/065,043, clinical trial registry, Indian Council of Medical Research) to assess its safety and efficacy in patients with hospital-acquired/ventilator-associated pneumonia (HAP/VAP), complicated intra-abdominal infections (cIAI) and cUTI, caused by meropenem-resistant Gram-negative bacteria. With regard to clinical experience, there are four case reports globally on the successful use of FEP/ZID for infections caused by DTR PsA. These include disseminated infection (bacteremia, pneumonia and ecthyma gangrenosum) in the background of hematological malignancy [6], sino-pulmonary infection & SBO in a renal transplant recipient [7], cIAI with VAP following bariatric surgery [8] and burns injury with secondary bacteremia, parapneumonic effusion and endophthalmitis [9]. All reported patients had successful outcomes.

Encouraged by the findings of these studies and with a dearth of safe and effective therapeutic options, we chose FEP/ZID for our patients with deep-seated infections caused by XDR PsA. Four cases involved osteoarticular disease (SBO and septic arthritis) while the fifth was an endovascular graft site infection. Presumably, all cases were complicated by significant biofilm production. Despite this, we observed a satisfactory and sustained clinico-radiological improvement in all our patients. This is likely due to its excellent biofilm eradication activity [19] and the effect of the BL-BLE combination wherein zidebactam lowers the minimum time required above minimum inhibitory concentration (% fT>MIC) for cefepime [5] and thus imparts clinical efficacy even at target sites with limited drug exposure. It has a remarkable safety profile, even on prolonged use (up to 6 wk). Larger clinical trials in patients with diverse clinical syndromes caused by varied resistotypes and comparison with the currently used standard of care would further strengthen its role as a vital treatment option for XDR Gram-negative infections.

5. Conclusion

In summary, we report the use of FEP/ZID for the successful management of five XDR PsA-related deep-seated infections complicated by biofilm formation. Similar to previously published case reports, our experience supports its safety and efficacy profile. These findings are particularly relevant in settings where antimicrobial resistance in PsA is concurrently mediated by an array of enzymatic and non-enzymatic mechanisms. Future clinical trials and compassionate use experiences are needed to further characterise its clinical utility.

Funding: None.

Declaration of competing interests: None declared.

Ethical approval: Institutional Review Board, Silver (Research and Ethics Committee) of the Christian Medical College, Vellore, Tamil Nadu, India. Reference number: IRB Min. No. 0,424,159.

Informed consent: Written informed consent was taken from each of the patients or their legal guardians/next of kin in accordance with the national legislation and the institutional requirements.

Data availability statement: The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2025.07.009.

References

- [1] GBD 2021 Antimicrobial Resistance Collaborators. Global burden of bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050. *Lancet* 2024;404(10459):1199–226 Epub 2024 Sep 16. PMID: 39299261; PMCID: PMC11718157. doi:10.1016/S0140-6736(24)01867-1.
- [2] WHO bacterial priority pathogens list, 2024: bacterial pathogens of public health importance to guide research, development and strategies to prevent and control antimicrobial resistance. [cited 2024 Jun 2]. Available from: <https://www.who.int/publications-detail-redirect/9789240093461>.
- [3] Cosentino F, Viale P, Giannella M. MDR/XDR/PDR or DTR? Which definition best fits the resistance profile of *Pseudomonas aeruginosa*? *Curr Opin Infect Dis* 2023;36(6):564–71. [cited 2024 Apr 10] Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10836784/>.
- [4] Yahav D, Giske CG, Grāmatniece A, Abodakpi H, Tam VH, Leibovici L. New β -lactam- β -lactamase inhibitor combinations. *Clin Microbiol Rev* 2020;34(1):e00115–20.
- [5] Lepak AJ, Zhao M, Andes DR. WCK 5222 (Cefepime-Zidebactam) pharmacodynamic target analysis against metallo- β -lactamase-producing Enterobacteriaceae in the neutropenic mouse pneumonia model. *Antimicrob Agents Chemother* 2019;63(12). [cited 2024 Oct 1] e01648–19 Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6879221/>.
- [6] Tirlangi PK, Wanve BS, Dubbudu RR, Yadav BS, Kumar LS, Gupta A, et al. Successful use of Cefepime-Zidebactam (WCK 5222) as a salvage therapy for the treatment of disseminated extensively drug-resistant New Delhi metallo- β -lactamase-producing *Pseudomonas aeruginosa* infection in an adult patient with acute T-cell leukemia. *Antimicrob Agents Chemother* 2023;67(8):e0050023.
- [7] Soman R, Sirsat R, Sunavala A, Punatar N, Mehta J, Rodrigues C, Veeraraghavan B. Successful treatment of sino-pulmonary infection & skull base osteomyelitis caused by New Delhi metallo- β -lactamase-producing *Pseudomonas aeruginosa* in a renal transplant recipient by using an investigational antibiotic cefepime/zidebactam (WCK 5222). *Eur J Clin Microbiol Infect Dis* 2024 Epub ahead of print. PMID: 38416290. doi:10.1007/s10096-024-04791-1.
- [8] Dubey D, Roy M, Shah TH, Bano N, Kulshrestha V, Mitra S, et al. Compassionate use of a novel β -lactam enhancer-based investigational antibiotic cefepime/zidebactam (WCK 5222) for the treatment of extensively-drug-resistant NDM-expressing *Pseudomonas aeruginosa* infection in an intra-abdominal infection-induced sepsis patient: a case report. *Ann Clin Microbiol Antimicrob* 2023;22(1):55.
- [9] Sebastian A, Naik S, Varma M, Sreekumar NC, Thomas J, Rao T, et al. Difficult to treat *Pseudomonas*: successful salvage therapy with cefepime-zidebactam. *Trans R Soc Trop Med Hyg* 2024;118(9):632–4 [cited 2024 Oct 1]. doi:10.1093/trstmh/trae049.
- [10] CLSI Performance standards for antimicrobial susceptibility testing. 34th ed. USA: Clinical and Laboratory Standards Institute; 2024.
- [11] Tamma PD, Aitken SL, Bonomo RA, Mathers AJ, van Duin D, Clancy CJ. Infectious Diseases Society of America guidance on the treatment of extended-spectrum β -lactamase producing enterobacterales (ESBL-E), carbapenem-resistant enterobacterales (CRE), and *Pseudomonas aeruginosa* with difficult-to-treat resistance (DTR-P. aeruginosa). *Clin Infect Dis* 2021;72(7):e169–83 [cited 2024 Jun 10]. doi:10.1093/cid/ciaa1478.
- [12] Lee YL, Ko WC, Hsueh PR. Geographic patterns of carbapenem-resistant *Pseudomonas aeruginosa* in the Asia-Pacific Region: results from the Antimicrobial Testing Leadership and Surveillance (ATLAS) Program, 2015–2019. *Antimicrob Agents Chemother* 2022;66(2):e02000–21. [cited 2024 Jun 10] Available from: <https://journals.asm.org/doi/10.1128/aac.02000-21>.
- [13] Karakostas S, Rousaki M, Kritsotakis EL. Cefiderocol: systematic review of mechanisms of resistance, heteroresistance and In vivo emergence of resistance. *Antibiotics* 2022;11(6):723. [cited 2024 Jun 11] Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9220290/>.
- [14] Wockhardt Ltd; A randomized, double-blind, placebo-controlled study to evaluate the safety, tolerability and pharmacokinetics of multiple escalating doses of intravenous zidebactam in healthy adult Human subjects; ClinicalTrials.gov identifier: NCT02674347; Updated : 2016; Accessed 02.10.24: <https://clinicaltrials.gov/study/NCT02674347?term=NCT02674347&rank=1>.
- [15] Wockhardt Ltd; A randomized, double-blind, placebo-controlled study to evaluate the safety, tolerability and pharmacokinetics of multiple escalating doses of intravenous WCK 5222 (Zidebactam and Cefepime) in Healthy adult human subjects; clinicaltrials.gov identifier: NCT02707107; 2016; Accessed 02.10.24 <https://clinicaltrials.gov/study/NCT02707107?term=NCT02707107&rank=1>.
- [16] Wockhardt Ltd; A randomized, double-blind, double-dummy, placebo- and positive-controlled, crossover study to evaluate the effect of WCK 5222 on the QT/QTc interval in healthy volunteers; ClinicalTrials.gov Identifier: NCT03554304; 2018; Accessed 02.10.24: <https://clinicaltrials.gov/study/NCT03554304?term=NCT03554304&rank=1>.
- [17] Wockhardt Ltd; A phase 1, open-label, single-dose study to investigate the pharmacokinetics of intravenous WCK 5222 (FEP-ZID) in patients with renal impairment; ClinicalTrials.gov Identifier : NCT02942810;

- 2018; Accessed 02.10.24; <https://clinicaltrials.gov/study/NCT02942810?intr=FEP-ZID&rank=3>.
- [18] Wockhardt Ltd; A phase 3, randomized, double-blind, multicenter, comparative study to determine the efficacy and safety of Cefepime-zidebactam vs. Meropenem in the treatment of complicated urinary tract infection or acute pyelonephritis in adults; ClinicalTrials.gov Identifier: NCT04979806; 2024; Accessed 02.10.24 <https://clinicaltrials.gov/study/NCT04979806?term=NCT04979806&rank=1>.
- [19] S Palwe, P Joshi, H Periasamy and S Bhagwat; Biofilm eradication activity of Zidebactam/Cefepime (WCK 5222, ZAYNICH), Assessed against Multi-Drug Resistant (MDR) *Pseudomonas aeruginosa* (PSA) Biofilms; CIDSCON (Conference of Clinical Infectious Diseases Society), August 15th to 18th 2024, Hyderabad, India.